

Investigating the Relationship Between Red Wine Quality and Key Chemical Properties: pH and Alcohol Content

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INTRODUCTION

Motivation

Our project focuses on exploring two potential chemical factors that contribute to the final perceived quality of red wines: alcohol percentage & pH. Understanding how they impact the perceived quality of red wines will help production plants identify what chemical factors to optimize and perfect. It will also help retailers set more accurate prices based on predicted wine quality of the *Vinho Verde* red wines. Lastly, analyzing the relationship between chemical factors and quality of red wine helps general consumers, like ourselves, make better decisions when buying red wine.

Background

Wine has become increasingly popular beyond luxury markets, and for major exporters like Portugal, ensuring quality is essential for both consumer trust and economic growth. Since taste is difficult to measure and not properly understood, wine quality tests usually combine chemical tests (such as acidity, alcohol, and pH) with expert sensory evaluations. The alcohol content and pH are important parameters that shape the sensory and chemical quality of red wines. Alcohol percentage influences qualities such as body, warmth, and aroma release, but excessive levels can create imbalance and influence varietal character. The pH governs acidity perception, color stability, tannin structure, and microbial safety, with lower pH enhancing freshness and long-term stability. High alcohol content with high pH produces heavy, unstable wines, while balanced values results in vibrancy and harmony. Exploring the links between these chemical factors and perceived quality can help winemakers refine their processes and assist retailers in identifying higher-quality wines.

Data Background

Our population of interest is all bottles of *Vinho Verde* red wines produced by production plants located in the northwestern regions of Portugal. We used the 'winequality-red.csv' dataset from Kaggle in our analysis. This sample of 1599 red wines was collected over a period of 3 years between May 2004 and February 2007 from production plants located in the northwestern region of Portugal. For each bottle of red wine produced, the dataset has recorded information on: Fixed acidity, Volatile acidity, Citric acid, Residual sugar, Chlorides, Free sulfur dioxide, Total sulfur dioxide, Density, pH, Sulfates, Alcohol, and Quality. The chemical factors were measured directly during the production process, and the quality score was determined by three independent physicochemical assessors.

Guiding Question

How does the alcohol percentage and final pH level contribute to the quality score of *Vinho Verde* red wines?

The two variables of focus were percent alcohol content and pH level due to their easy comprehension. They are accessible by broader audiences and non-experts, as opposed to, for example, 'total sulfur dioxide', which is unintelligible for the general populace.

METHODS

Our primary variables of interest were alcohol percentage, pH, and quality score. Alcohol percentage and pH were continuous numeric variables, while quality score was an ordinal categorical variable with values from 0 to 10. For our analysis, we split the red wines into a high quality and a low quality category based off of the quality score. Wines with a quality score of 6 and above were sorted into the high quality group and wines with a quality score of 5 and below were sorted into the low quality group. The alpha value for our hypothesis tests is 0.05.

Question 1: Do high quality *Vinho Verde* red wines have a significantly higher percent alcohol content compared to low quality red wines?

We conducted a two sample independent t-test to check whether the average percent alcohol content for red wines in the high quality category was significantly higher than the average percent alcohol content for red wines in the low quality category. We assumed normality through the Central Limit Theorem since the sample sizes in both high quality and low quality categories are well above $n=25$. As a supplement to our two sample independent t-test, we also conducted a permutation test for the difference between average percent alcohol content between the high quality group and low quality group at 4999 iterations.

Finally we were also interested in the 95% confidence interval for the difference in mean percent alcohol content between high quality and low quality red wines. We computed this statistic through a 95% confidence interval using a student t distribution.

Question 2: Is the final pH level of *Vinho Verde* red wines significantly associated with its perceived quality?

We conducted a Chi-squared test of independence to check whether the quality category of the red wines was independent of its pH level. Red wines were sorted into 3 different pH categories based on current industry standards and they were also sorted into high quality and low quality categories based on their quality score resulting in a 2x3 crosstable. Since the industry cutoffs for the pH values may result in highly unbalanced bins, we supplemented our analysis using quartile values for the pH category cutoffs resulting in a 2x4 crosstable.

DATA EXPLORATION

Data Wrangling and Summary Statistics

We created the **quality.cat** variable to categorize the red wines based on quality score. Red wines with a quality score of 6 and above were assigned “High”, otherwise they were assigned “Low”.

```
> favstats(~alcohol|quality.cat, data=wine.df)
```

	quality.cat	min	Q1	median	Q3	max	mean	sd	n	missing
1	High	8.4	10.0	10.8	11.7	14.0	10.855029	1.1061093	855	0
2	Low	8.4	9.4	9.7	10.3	14.9	9.926478	0.7580065	744	0

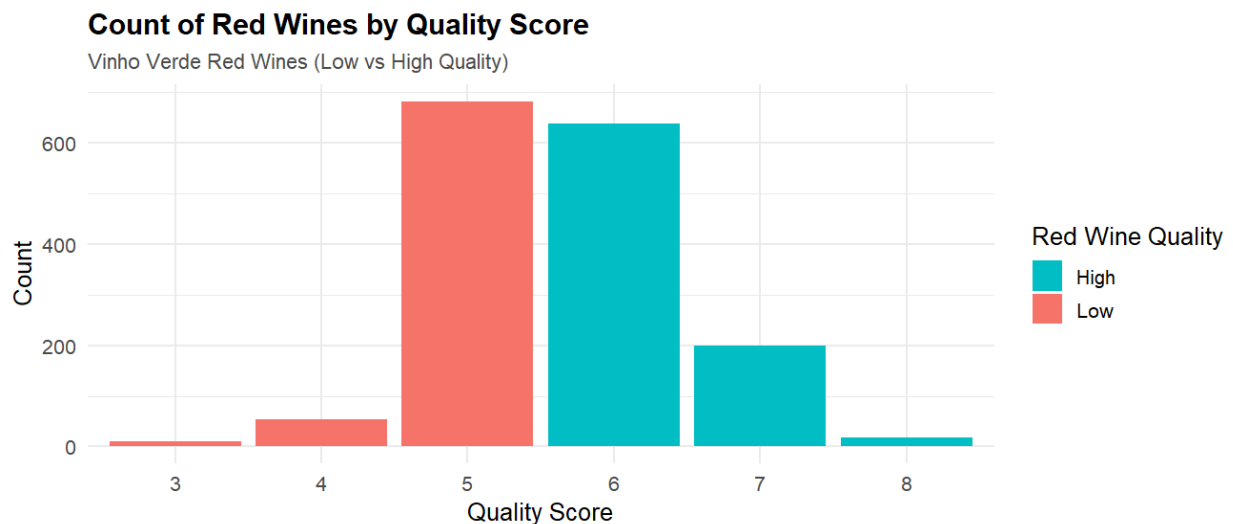
The summary statistics of the % alcohol content of the red wines showed that the mean % alcohol content was different between the two quality groups by ~0.93%. For the low quality red wine group the average percent alcohol content was $9.93\% \pm 0.76\%$ and for the high quality group, it was $10.86\% \pm 1.10\%$. There were no missing values in percent alcohol content for either quality groups.

```
> favstats(wine.df$pH)
```

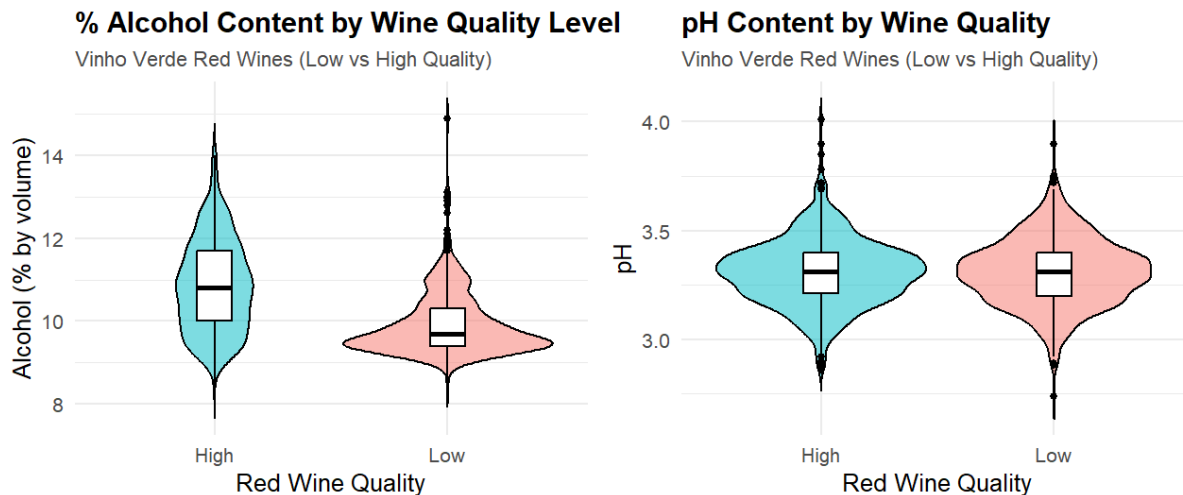
	min	Q1	median	Q3	max	mean	sd	n	missing
	2.74	3.21	3.31	3.4	4.01	3.311113	0.1543865	1599	0

The summary statistics for the pH levels in the red wine dataset showed that the mean pH level was 3.31 ± 0.15 . The most acidic bottle in this sample has a pH of 2.74 while the most basic has a pH of 4.01, the median of the pH levels was 3.31 which was very close to the mean. There were no missing values in the pH level variable.

Data Visualization



The distribution of red wine quality was highly centralized around the ratings of 5 and 6 which together accounts for the majority of the samples included in this dataset. While the quality score is 0-10, the range for this dataset was 3-8. There were very few wines which had extreme quality scores of 3-4 or 7-8. The division between the low quality ($n=744$) and high quality ($n=855$) groups was relatively balanced, suggesting that the dataset captured a representative mix of both high quality and low quality red wines. This balance ensures that our analysis comparing differences between the high quality and low quality groups of red wines remains relatively unbiased.



The violin plot on the left illustrates the distribution of alcohol content in *Vinho Verde* red wines, comparing low-quality and high-quality categories. The high-quality wines seem to have higher alcohol content, with a median appearing higher than the low-quality wines. The distribution for high-quality wines was also more spread out, indicating greater variability in alcohol levels. The wider areas of each violin represent where more data points are concentrated, showing that alcohol percentage varied more widely among high-quality wines. Overall, the plot suggests that higher alcohol content is associated with higher wine quality.

The violin plot on the right displays the distribution of pH levels in *Vinho Verde* red wines, comparing low-quality and high-quality categories. Both groups visually show similar median pH values, potentially indicating that pH does not vary much between wine quality levels. The distributions were also similar in shape and spread, suggesting that acidity was not a strong distinguishing factor between low- and high-quality wines. Overall, wine quality visually appears to be relatively independent of pH content in this dataset with needed statistical analysis for confirmation.

HYPOTHESIS TESTING

Question 1: Do high quality *Vinho Verde* red wines have a significantly higher percentage alcohol content compared to low quality red wines?

Statistical Hypotheses

$$H_0: \mu_{\%alcohol}^{High\ quality} \leq \mu_{\%alcohol}^{Low\ quality}$$

$$H_A: \mu_{\%alcohol}^{High\ quality} > \mu_{\%alcohol}^{Low\ quality}$$

Assumption Check

The sample size in the high quality wine group is $n=855$ and the sample size in the low quality wine group is $n=744$, both of which are greater than $n=25$. Therefore we can assume that the distribution of percentage alcohol content within each quality group is approximately normal through the CLT.

Statistical Analysis

```
> t.test(~alcohol|quality.cat, data=wine.df, alternative="greater")
```

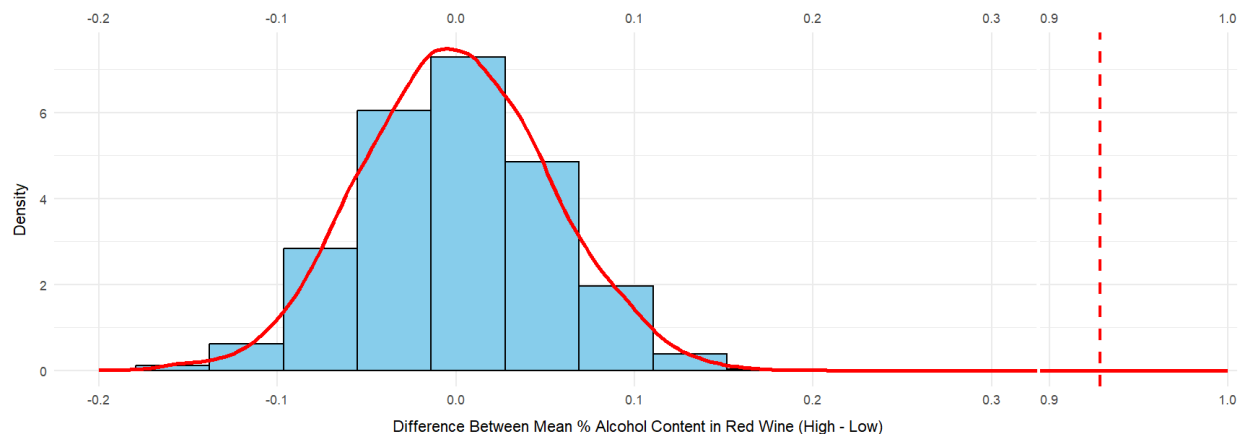
Welch Two Sample t-test

```
data: alcohol by quality.cat  
t = 19.782, df = 1516.8, p-value < 2.2e-16
```

The t statistic of the independent two sample t-test is 19.782 and the p-value is $< 2.2e-16$. The 95% confidence interval for the difference in % alcohol content between high and low quality groups is 0.84 and 1.02. We reject the null hypothesis and conclude that high quality red wines have significantly higher percentage alcohol content compared to low quality red wines.

Permutation Distribution of Mean Alcohol Differences

High-quality vs Low-quality Vinho Verde Red Wines



The figure above shows the 4999 iteration permutation distribution with the threshold value as the red dashed line. The threshold value for the permutation test is 0.9286 which is greater than all values in the 4999 iteration permutation distribution. The empirical p-value is $2e-4$, which is smaller than the alpha of 0.05. Therefore, we reject the null hypothesis and conclude that high quality red wines have a significantly higher percentage alcohol content compared to low quality red wines.

```
> t.test(~alcohol|quality.cat, data=wine.df)$conf  
[1] 0.836479 1.020622
```

Based on the 95% confidence interval calculated through the student's t distribution, we can be 95% confident that the true difference in percent alcohol content between high quality and low quality red wines is between 0.8365 and 1.0206%.

Question 2: Is the final pH level of *Vinho Verde* red wines significantly associated with its perceived quality?

Statistical Hypotheses

H_0 : The quality of the red wine is independent from its pH level

H_A : The quality of the red wine is not independent from its pH level

pH Level Cutoffs

Chemical based cutoffs:

According to standard wine literature, the typical pH range for red wines lies between 3.30 and 3.60, with wines below pH 3.20 considered to have high acidity and wines above 3.50 considered low in acidity. Based on these references, we defined 3 pH categories to aid interpretation. < 3.20 for high acidity, 3.20-3.50 for moderate acidity, and > 3.50 for low acidity. This value is recorded under the **pH.cat.chem** variable.

Quartile cutoffs:

To account for the potential imbalance caused by chemical based cutoffs, we ran a concurrent analysis where pH was divided into 4 categories based on the quartile summary stats shown under the Data Exploration section. 2.74-3.21 for high acidity, 3.21-3.31 for high-moderate acidity, 3.31-3.40 for low-moderate acidity, and 3.40-4.01 for low acidity. This value is recorded under the **pH.cat.bal** variable.

Crosstables will be created through using one of the above pH value cutoffs with the red wine quality category variable (**quality.cat**).

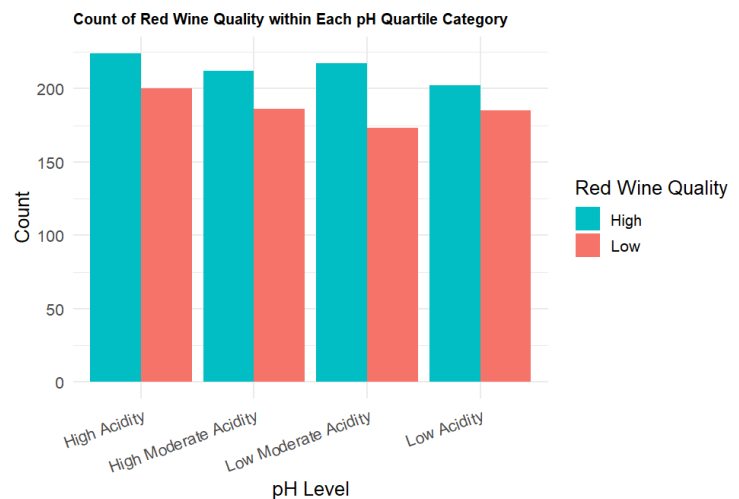
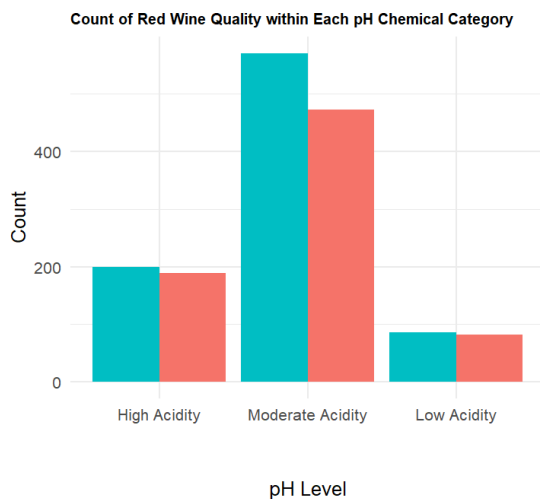
Assumption Check

```
> chem.ct
```

	High Acidity	Moderate Acidity	Low Acidity
Low Quality	189	473	82
High Quality	199	570	86

```
> bal.ct
```

	High Acidity	High Moderate Acidity	Low Moderate Acidity	Low Acidity
Low Quality	200	186	173	185
High Quality	224	212	217	202



The crosstables and barplots for the chemical pH category and quartile pH categories are shown above. Each category has at least 5 counts, thus we can proceed with the Chi-squared test of independence.

Statistical Analysis

```
> xchisq.test(chem.ct, correct=F)

Pearson's Chi-squared test
```

```
data: x
X-squared = 1.6767, df = 2, p-value = 0.4324
```

The Chi-squared statistic is 1.6767 with a p-value of 0.4324 for the chemical pH cutoffs.

```
> xchisq.test(bal.ct, correct=F)

Pearson's Chi-squared test
```

```
data: x
X-squared = 1.0676, df = 3, p-value = 0.7849
```

The Chi-squared statistic is 1.0676 with a p-value of 0.7849 for the quartile pH cutoffs.

Both p-values are greater than the set alpha level of 0.05. Therefore, we fail to reject our null hypothesis and conclude that the perceived quality of *Vinho Verde* red wines are not significantly dependent on their final pH levels.

CONCLUSION

Result Interpretation

Our analyses addressed two key questions regarding *Vinho Verde* red wine quality. First, we found compelling statistical evidence that higher-quality wines tend to have higher alcohol content. The observed mean difference of ~0.93% alcohol was both statistically significant and practically meaningful, suggesting that alcohol concentration may play a key role in shaping the sensory attributes associated with higher-rated wines (i.e. body, aroma intensity, and perceived balance).

In contrast, wine pH level showed no significant association with perceived quality under either chemical or quartile-based categorizations. This independence implies that within the typical pH range observed for *Vinho Verde* reds, acidity does not appear to influence sensory evaluation in a measurable way.

In all, these results suggest that while alcohol content may serve as a reliable indicator of higher quality in *Vinho Verde* red wines, pH alone is an insignificant predictor.

Future Direction

Future studies can expand on this work by exploring a wider range of chemical and sensory factors influencing red wine quality. Other variables such as volatile acidity, sulphates, and sulphur dioxide content could be considered to describe interactively how multiple parameters influence wine quality in a bigger picture. Using other techniques, such as multiple linear regression, principal component analysis, or machine learning algorithms, it will be feasible to identify complex and non-linear relationships between different chemical factors in their cumulative determination of perceived quality.

Furthermore, providing more accurate sensory criteria (i.e., aroma intensity, tannin structure, and color) for human evaluators would likely yield better analysis and provide deeper insight into overall wine quality.

Expanding the dataset to other regions in Portugal or to more recent vintages may show whether these associations are affected by temporal or climatic changes. It may be worth carrying out experimental studies whereby fermentation conditions, pH, and alcohol levels are all manipulated to give more robust causal evidence about how these factors influence flavor and stability. The development of predictive models for wine quality from chemical composition will assist manufacturers and retailers in quality control, pricing, and product design in line with consumer trends.

Another potential direction for future research is the biological aspect of wine fermentation: studying how to modify yeast strains to reach the optimal level of alcohol and the desired flavor profile. Through genetic and metabolic engineering, selective breeding, mutagenesis, and adaptive evolution, winemakers could create yeast strains with better fermentation efficiency or more regulated ethanol and glycerol formation pathways. These methods can affect the flux of the metabolic processes of glycerol and ethanol during fermentation, where the winemaker can not only control ethanol concentration but also control the texture, sweetness, and aromas in the final product. In addition to the possible goal of optimizing yeast strains, utilizing naturally occurring yeast strains or simply isolating and selecting yeasts that perform well given the fermentation conditions, may help improve not only sensory quality but also the consistency across vintage years and wines within a specific geographic area.

REFERENCES

- Atlas Scientific. (2023, July 7). *The importance of pH in wine-making*. https://atlas-scientific.com/blog/the-importance-of-ph-in-wine-making/?srsltid=AfmBOOp9MYjk5_0E7wn2nD5DSpdbH1rW2rsacn4zWHLYUgoJiSE7\~lx4
- Boulton, R. (1980). The general relationship between potassium, sodium and pH in grape juice and wine. *American Journal of Enology and Viticulture*, 31(2), 182–186. <https://doi.org/10.5344/ajev.1980.31.2.182>
- Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems*, 47(4), 547–553. <https://doi.org/10.1016/j.dss.2009.05.016>
- Giuffrida de Esteban, M. L., Ubeda, C., Heredia, F. J., Catania, A. A., Assof, M. V., Fanzone, M. L., & Jofre, V. P. (2019). Impact of closure type and storage temperature on chemical and sensory composition of Malbec wines (Mendoza, Argentina) during aging in bottle. *Food Research International*, 125, 108553. <https://doi.org/10.1016/j.foodres.2019.108553>
- Keller, M. (2015). Botany and anatomy. *The Science of Grapevines*, 1–57. <https://doi.org/10.1016/b978-0-12-419987-3.00001-7>
- UCI Machine Learning. (2018). *Red Wine Quality*. www.kaggle.com. <https://www.kaggle.com/datasets/uciml/red-wine-quality-cortez-et-al-2009>
- Waterhouse, A. L., Sacks, G. L., & Jeffery, D. W. (2016). *Understanding Wine Chemistry*. <https://doi.org/10.1002/9781118730720>

APPENDIX

R Code

```
#File Startup, list of libraries used in analysis and visualization
rm(list=ls(all=TRUE))
library(ggplot2)
library(mosaic)
library(patchwork)
library(ggbreak)

###INTRODUCTION
##Data Wrangling and Summary Statistics
#Read the dataset into R
wine.df<-read.csv("C:\\Users\\yuhao\\Desktop\\Uni\\DATA
602\\winequality-red.csv")

#Create and assign values to the quality.cat variable based on wine quality
score
```

```

quality.cat<-c()
for (i in wine.df$quality){
  if (i <= 5){quality.cat<-c(quality.cat, "Low")}
  else {quality.cat<-c(quality.cat, "High")}
}

#Add the quality.cat variable into the wine data frame
wine.df<-data.frame(wine.df, quality.cat)

#Generate summary statistics for % alcohol content based on quality group
favstats(~alcohol|quality.cat, data=wine.df)

#Generates summary statistics for pH level variable
favstats(wine.df$pH)

##Data Visualization
#Barplot showing the distribution of quality scores
wine.df$quality1<-factor(wine.df$quality)

quality.plot<-ggplot(wine.df, aes(x=quality1, fill=quality.cat))+
  geom_bar()+
  scale_fill_manual(values=c("Low"="#F8766D", "High"="#00BFC4"))+
  theme_minimal(base_size=13)+
  labs(
    title="Count of Red Wines by Quality Score",
    subtitle="Vinho Verde Red Wines (Low vs High Quality)",
    x="Quality Score",
    y="Count",
    fill="Quality")+
  theme(
    plot.title=element_text(face="bold", size=15),
    plot.subtitle=element_text(size=11, color="gray30"))
quality.plot$labels$fill<-"Red Wine Quality"
quality.plot

#Violin plot for the distribution of % alcohol content between high and low
quality groups
alcohol.plot<-ggplot(wine.df, aes(x=quality.cat, y=alcohol,
fill=quality.cat))+
  geom_violin(trim=FALSE, alpha=0.5, color="black")+
  geom_boxplot(width=0.15, outlier.shape=16, outlier.size=1.5, color="black",
fill="white")+
  scale_fill_manual(values=c("Low"="#F8766D", "High"="#00BFC4"))+
  theme_minimal(base_size=13)+
  labs(
    title="% Alcohol Content by Wine Quality Level",
    subtitle="Vinho Verde Red Wines (Low vs High Quality)",
    x="Red Wine Quality",
    y="Alcohol (% by volume)",
    fill="Quality")+

```

```

theme(
  plot.title=element_text(face="bold", size=10),
  plot.subtitle=element_text(size=11, color="gray30"),
  legend.position="none")

#Violin plot for the distribution of pH level between high and low quality
groups.
pH.plot<-ggplot(wine.df, aes(x=quality.cat, y=pH, fill=quality.cat))+
  geom_violin(trim=FALSE, alpha=0.5, color="black")+
  geom_boxplot(width=0.15, outlier.shape=16, outlier.size=1.5, color="black",
fill="white")+
  scale_fill_manual(values=c("Low"="#F8766D", "High"="#00BFC4"))+
  theme_minimal(base_size=13)+
  labs(
    title="pH Content by Wine Quality",
    subtitle="Vinho Verde Red Wines (Low vs High Quality)",
    x="Red Wine Quality",
    y="pH",
    fill="Quality")+
  theme(
    plot.title=element_text(face="bold", size=10),
    plot.subtitle=element_text(size=11, color="gray30"),
    legend.position="none")

#Combines the two plots above into one graphic
alcohol.plot+pH.plot

####HYPOTHESIS TESTING Q1
##Statistical Analysis
#Rightsided independent two sample t-test
t.test(~alcohol|quality.cat, data=wine.df, alternative="greater")

#Rightsided permutation test using 4999 iterations
threshold<-mean(subset(wine.df,
quality.cat=="High")$alcohol)-mean(subset(wine.df,
quality.cat=="Low")$alcohol)
threshold

n<-length(wine.df$alcohol)
n.high<-length(subset(wine.df, quality.cat=="High")$alcohol)
n.low<-length(subset(wine.df, quality.cat=="Low")$alcohol)

set.seed(1111)
N=4999
permvector=numeric(N)
for (i in 1:N)
{
  index=sample(n, n.high, replace = FALSE)
  permvector[i] = mean(wine.df$alcohol[index])-mean(wine.df$alcohol[-index])
}

```

```

(sum(permvector >= threshold)+1)/(N+1)

#Histogram for the distribution of mean differences in % alcohol content
#Adds scale breaks on the x-axis to show threshold value
permvector.df = data.frame(permvector)
perm.plot<-ggplot(permvector.df, aes(x=permvector))+
  geom_histogram(aes(y=after_stat(density)), col = "black", fill =
"skyblue")+
  scale_x_break(c(0.3, 0.9)) +
  geom_vline(xintercept = threshold, color = "red", linetype = "dashed", size
= 1)+
  geom_density(col = "red", size = 1.2)+
  theme_minimal()+
  labs(
    title = "Permutation Distribution of Mean Alcohol Differences",
    subtitle = "High-quality vs Low-quality Vinho Verde Red Wines",
    x = "Difference Between Mean % Alcohol Content in Red Wine (High - Low)",
    y = "Density"
  )+theme(plot.title=element_text(face="bold", size=15))
perm.plot + scale_x_continuous(
  limits = c(-0.2, 1),
  breaks = c(-0.2, -0.1, 0, 0.1, 0.2, 0.3, 0.9, 1.0),
  minor_breaks = NULL)

#95% confidence interval for difference in mean % alcohol content from
students t distribution
t.test(~alcohol|quality.cat, data=wine.df)$conf

####HYPOTHESIS TESTING Q2
##Data Wrangling and Assumption Check
#Created variable for chemical pH categories
pH.cat.chem<-c()
for (i in wine.df$pH){
  if (i <= 3.2){pH.cat.chem<-c(pH.cat.chem, 1)}
  else if (i > 3.2 & i <= 3.5){pH.cat.chem<-c(pH.cat.chem, 2)}
  else{pH.cat.chem<-c(pH.cat.chem, 3)}
}

#Created variable for quartile pH categories
pH.cat.bal<-c()
for (i in wine.df$pH){
  if (i <= 3.21){pH.cat.bal<-c(pH.cat.bal, 1)}
  else if (i > 3.21 & i <= 3.31){pH.cat.bal<-c(pH.cat.bal, 2)}
  else if (i > 3.31 & i <= 3.40){pH.cat.bal<-c(pH.cat.bal, 3)}
  else{pH.cat.bal<-c(pH.cat.bal, 4)}
}

#Add the chemical and quartile pH categories back into wine dataframe
wine.df<-data.frame(wine.df, pH.cat.chem, pH.cat.bal)

```

```

#Create crosstables for chemical pH categories
ct.chem<-c()
quality<-c("Low", "High")
for (i in quality){
  for (j in 1:3){
    ct.chem<-c(ct.chem,length(subset(wine.df, quality.cat==i &
pH.cat.chem==j)$quality))
  }
}

chem.ct<-rbind(c(189, 473, 82), c(199, 570, 86))
rownames(chem.ct)<-c("Low Quality", "High Quality")
colnames(chem.ct)<-c("High Acidity", "Moderate Acidity", "Low Acidity")

#Create crosstable for balanced pH categories
ct.bal<-c()
for (i in quality){
  for (j in 1:4){
    ct.bal<-c(ct.bal,length(subset(wine.df, quality.cat==i &
pH.cat.bal==j)$quality))
  }
}

bal.ct<-rbind(c(200, 186, 173, 185), c(224, 212, 217, 202))
rownames(bal.ct)<-c("Low Quality", "High Quality")
colnames(bal.ct)<-c("High Acidity", "High Moderate Acidity", "Low Moderate
Acidity", "Low Acidity")

#Create bar plot for chemical pH category crosstable
wine.df$pH.cat.chem1<-factor(wine.df$pH.cat.chem,levels=c(1, 2, 3),
                             labels=c("High Acidity", "Moderate Acidity",
"Low Acidity"))

chem.ct.plot<-ggplot(data=wine.df, aes(x=pH.cat.chem1, fill=quality.cat))+
  geom_bar(position="dodge")+
  scale_fill_manual(values=c("Low"="#F8766D", "High"="#00BFC4"))+
  theme_minimal(base_size=13)+
  labs(
    title="Count of Red Wine Quality within Each pH Chemical Category",
    x="pH Level",
    y="Count",)+
  theme(
    plot.title=element_text(face="bold", size=10),
    plot.subtitle=element_text(size=11, color="gray30"))
chem.ct.plot$labels$fill<-"Red Wine Quality"

wine.df$pH.cat.bal1<-factor(wine.df$pH.cat.bal,levels=c(1, 2, 3, 4),
                             labels=c("High Acidity", "High Moderate Acidity",
"Low Moderate Acidity", "Low Acidity"))

```

```

#Create barplot for quartile pH category crosstable
bal.ct.plot<-ggplot(data=wine.df, aes(x=pH.cat.bal1, fill=quality.cat))+
  geom_bar(position="dodge")+
  scale_fill_manual(values=c("Low"="#F8766D", "High"="#00BFC4"))+
  theme_minimal(base_size=13)+
  labs(
    title="Count of Red Wine Quality within Each pH Quartile Category",
    x="pH Level",
    y="Count",)+
  theme(
    axis.text.x=element_text(angle=20, hjust=1),
    plot.title=element_text(face="bold", size=10),
    plot.subtitle=element_text(size=11, color="gray30"))
bal.ct.plot$labels$fill<-"Red Wine Quality"

#Combine the above two plots into a single graphic and keeps only one legend
q2.plot<-chem.ct.plot+bal.ct.plot+plot_layout(guides = 'collect')
q2.plot

#Chi-squared test of independence based on chemical pH crosstable
xchisq.test(chem.ct, correct=F)

#Chi-squared test of independence based on quartile pH crosstable
xchisq.test(bal.ct, correct=F)

```